

DEPARTMENT OF BIO ENGINEERING

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Food Technology Engineering Bachelor Programs

Introduction to Degree

The curriculum is designed education in the principles of Bio Engineering and also development of communication, teamwork, leadership and entrepreneurship skills. Introduction to Degree

The bachelor program of Engineering in Food Technology and Engineering was established in February 2022. The Program focus on the most need of food technologist and engineer in food industry to support and push forward Cambodia's development on agro-industry sustainably. Program the address gaps in knowledge and skills and promotes high commitment to the development of agro-industry and food security. The program was developed with the participation of experts and researchers from Osaka Prefecture University. This program is designed to provide students with the knowledge, skills, and experience necessary for successful career in food technology in Cambodia as well as in Southeast Asia

Last semester Optional tracks



Coursework	is for students who would like to conduct coursework and final project.
Research Thesis	is for outstanding students who would like to conduct their research project under the supervision of the faculty members of the Department.

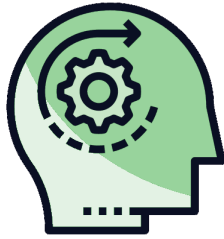
Extended Training

is designed for students who would like to participate in a long training period (for the entire semester) under a co-operative training program with companies or organizations.

TO IMPROVE STUDENT'S SOFT SKILL (COMMUNICATION, TEAMWORK AND LEADERSHIP), THE STUDENTS ARE REQUIRED TO CONDUCT A MINIMUM 6 WEEKS INTERNSHIP AT A STANDARD COMPANY OR ORGANIZATION DURING THEIR VACATION TIME OF THE THIRD YEAR OR BEFORE.

Prospects

The Department carries out research in which environmental impact and risk are assessed, monitored and/or address for enhancing environmental conservation and sustainability in Cambodia. Findings from the research can be used to facilitate recovering the environment and develop solutions to minimize the impacts on and risk to environmental resources in the country.



The Department provides its graduates a competitive edge in the labor market. We prepare high quality professionals to work as

Job Prospects

- Supervisor/ Production Planner
- QA and QC Specialist/ Food Safety specialist/Laboratory Manager
- Food Technologist/ R&D Chef
- Business Analyzer/ Stat-up

Evaluation & Graduation

Assessment of study will be performed at the end of each semester. Evaluation will be shown as the Grade Point Average (GPA) (4.00 scale) and the Cumulative Grade Point Average (4.00 scale).

Score Range	GPA	Grade Meaning
85–100	4.0	A = Excellent
80–84	3.5	B+ = Very Good
70–79	3.0	B = Good
65–69	2.5	C+ = Fairly Good
50–64	2.0	C = Fair
45–49	1.5	D = Poor
<40	0.0	E = Fail

Curriculum

The curriculum is designed education in the principles of Bio Engineering and also development of communication, teamwork, leadership and entrepreneurship skills.

L = Lecture, P = Practice, S = Self Study

Foundation

Subject	Credit	Hours (L - P - S)
SEMESTER I		
English Composition: (Critical thinking, reading and writing)	3.0	(3-0)
General Mathematics: Algebra	3.0	(3-0)
General Science 1: General Chemistry	4.0	(3-2)
General Physics	4.0	(3-0)
Fundamental Computer and Programming	3.0	(2-2)
Sociology: Digital skill and Problem Solving	3.0	(3-0)
Personal Development and Management	3.0	(3-0)
Introduction to Engineering	3.0	(3-0)
SEMESTER II		
English Composition: Communication skill	3.0	(3-0)
Statistics for Engineer	4.0	(3-1)
Biological and Environmental Engineering	3.0	(3-0)
Mathematics II (Calculus)	3.0	(3-0)
Khmer and the world culture	3.0	(3-0)
Computer Program Language	3.0	(3-0)
Data Sciences for Engineers	3.0	(3-0)
Engineering Drawing	3.0	(3-0)

Second Year

Subject	Credit	Hours (L - P - S)
SEMESTER I		
Introduction to Food Technology (Concept of engineering and technology)	3.0	(3-0)
Principle of Analytical Chemistry	3.0	(3-1)
Principle of physical chemistry	4.0	(3-0)
Organic chemistry	4.0	(3-1)
Introduction to Microbiology	4.0	(3-1)
Introduction to Economic	3.0	(3-1)

Subject	Credit	Hours (L - P - S)
Seminar in Food Technology	2.0	(2-0)
Food Processing 1	3.0	(3-0)
SEMESTER II		
Principle of instrumental analysis	4.0	(3-2)
Biochemistry	4.0	(3-1)
Food Chemistry	3.0	(3-0)
Food processing	4.0	(3-2)
Food properties	3.0	(3-0)
Food Microbiology	3.0	(3-0)
Introduction to Business	3.0	(3-0)
Free elective course	-	(-)

Third Year

Subject	Credit	Hours (L - P - S)
SEMESTER I		
Chemistry and Physical for food analysis	4.0	(3-1)
Food Nutrition	3.0	(3-0)
Food Engineering	4.0	(3-1)
Food Preservation and Packaging technique	3.0	(3-0)
Introduction to Enzymology in Food	3.0	(3-0)
Sensory Evaluation	4.0	(3-2)
Marketing and Management	2.0	(2-0)
SEMESTER II		
Design and Experimental planning food science	3.0	(3-0)
Food product development	3.0	(3-0)
Chemistry and technology of milk and dairy products	4.0	(3-2)
Startup business with food science and technology	2.0	(2-0)
Food process Engineering	4.0	(3-2)
Special Problems	2.0	(0-2)
Professional Experience	3.0	(0-135*)

Fourth Year

Subject	Credit	Hours (L - P - S)
SEMESTER I		
Food Toxicology and additive	4.0	(3-1)
Technology in Carbohydrate and Protein	4.0	(3-2)

Subject	Credit	Hours (L - P - S)
Food Law and Standard	3.0	(3-0)
Food Plant Management and Sanitation	3.0	(3-2)
Project Management in Food Industry	2.0	(2-0)
Principle of quality assurance and quality control in food	2.0	(2-0)
Food in multiculture	1.0	(3-0-6)
Internship (Soft Skill) (vacation time)	1.0	(1-0)
SEMESTER II		
Final project (industry training)	10.0	(2-18)
Thesis	10.0	(2-18)

About RUPP

The Royal University of Phnom Penh is Cambodia's oldest and largest public university, offering degrees in science, technology, humanities, and social sciences.

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